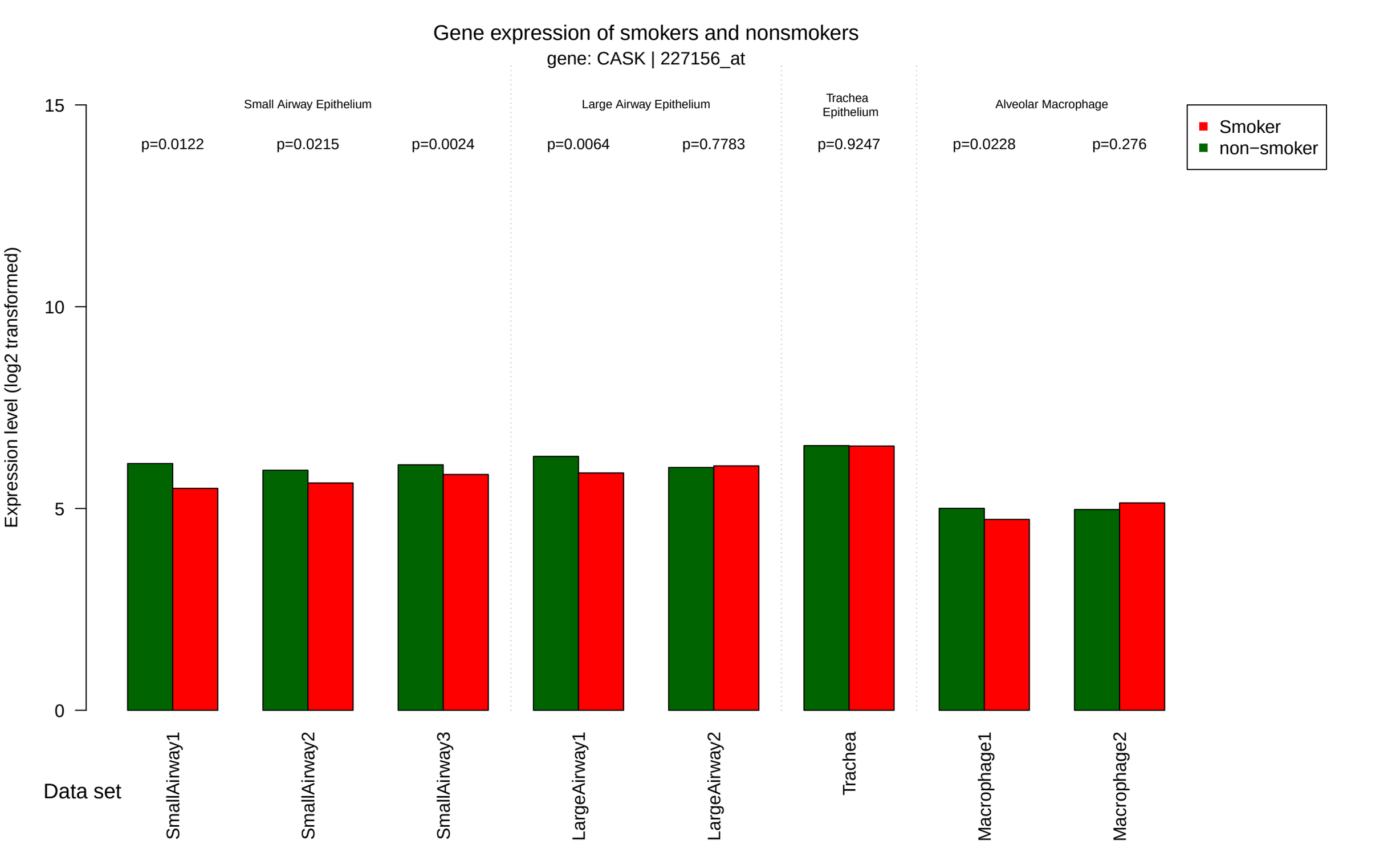
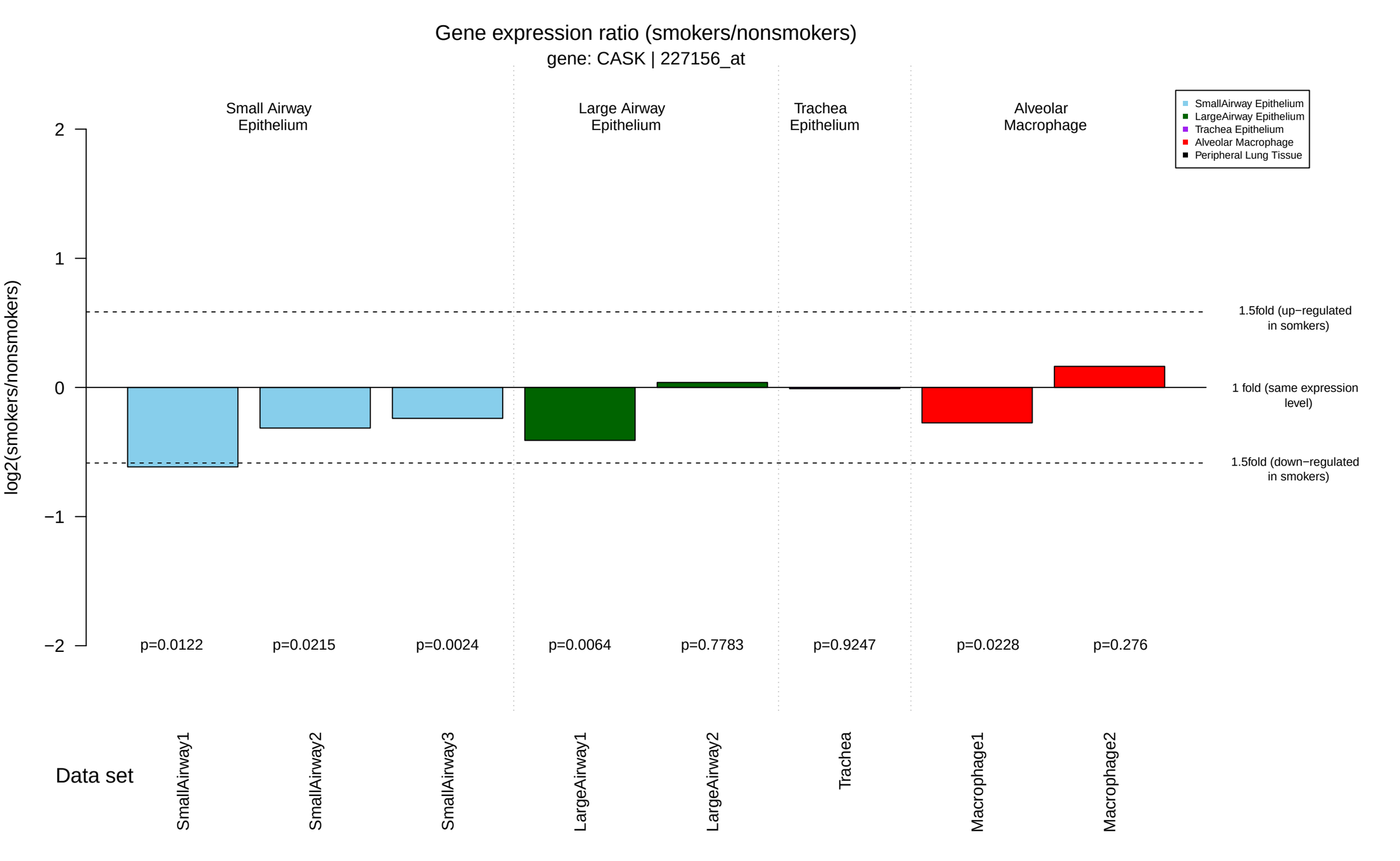
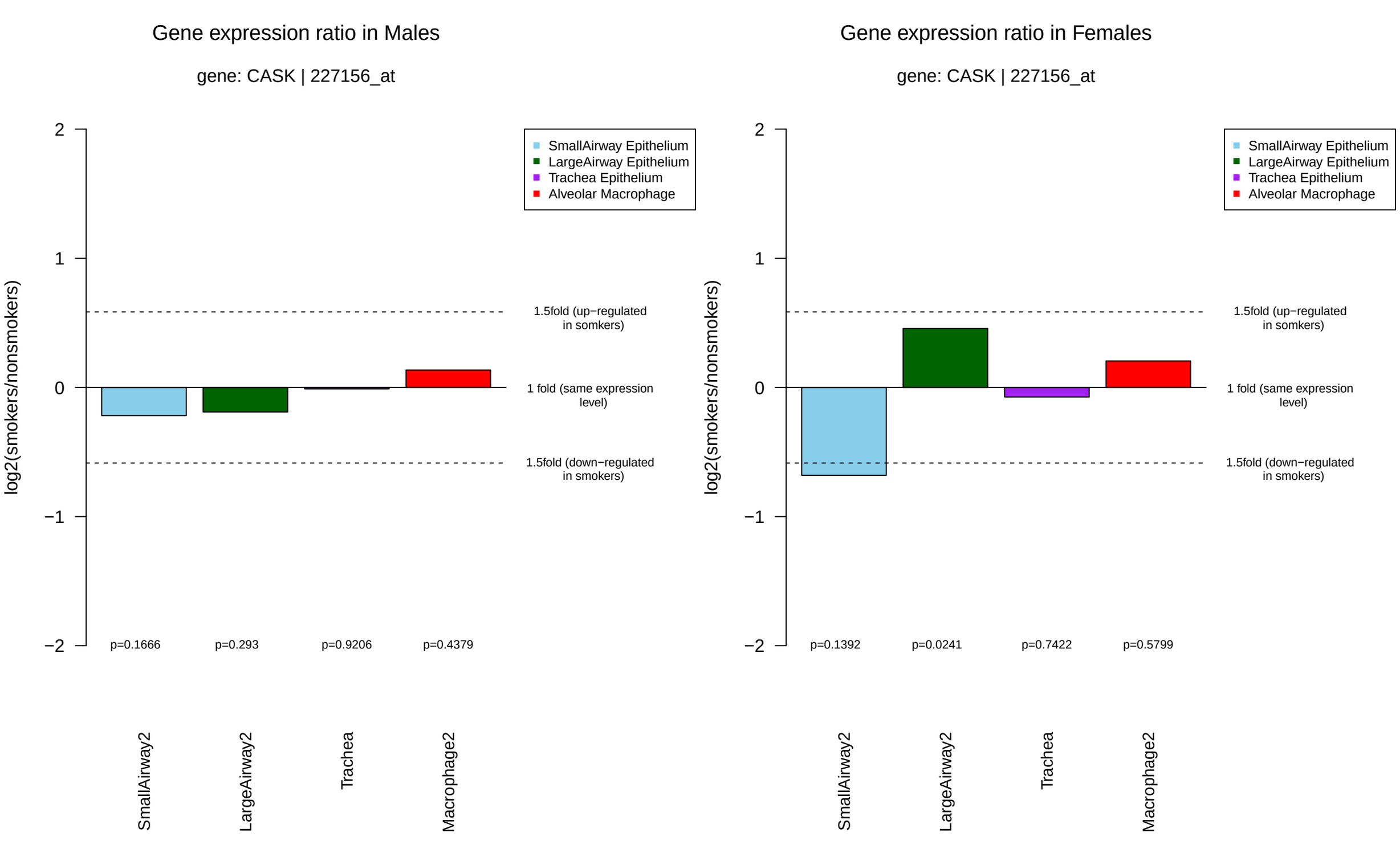
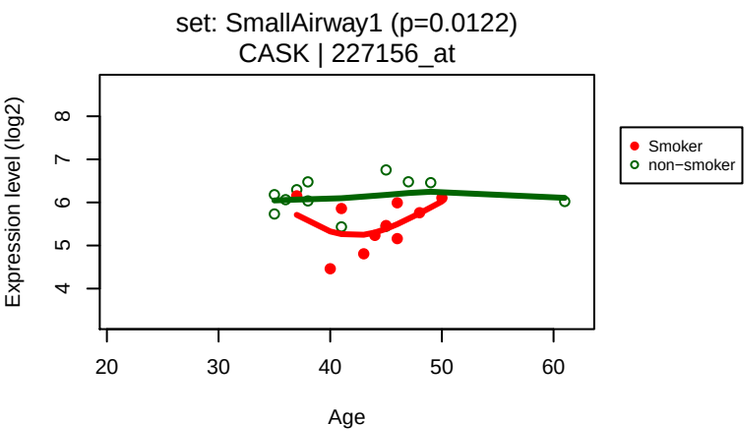
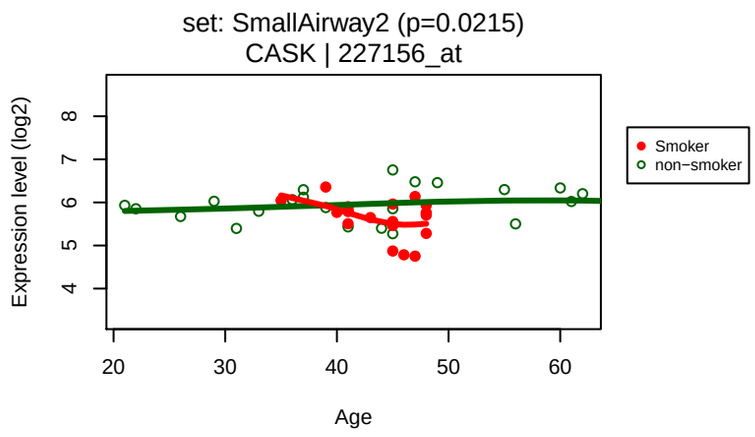
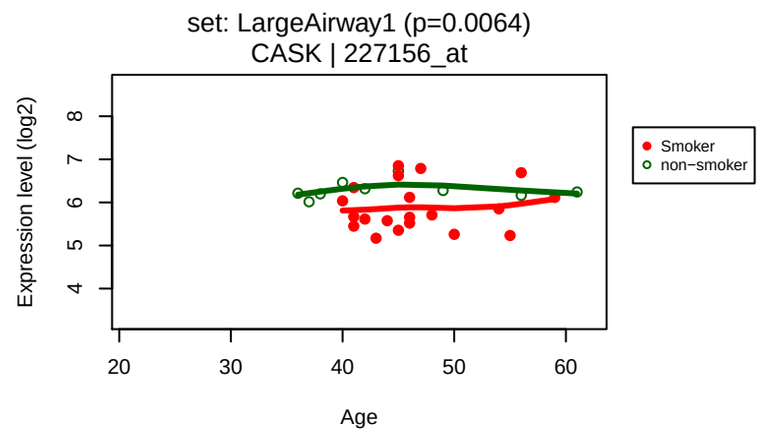

Gene expression of smokers and nonsmokers
gene: CASK | 227156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0122
p=0.0215
p=0.0024
p=0.0064
p=0.7783
p=0.9247
p=0.0228
p=0.276
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

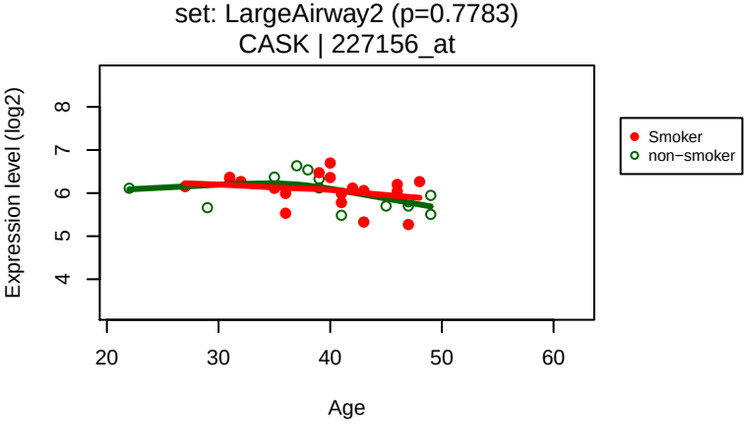

Gene expression ratio (smokers/nonsmokers)
gene: CASK | 227156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0122
p=0.0215
p=0.0024
p=0.0064
p=0.7783
p=0.9247
p=0.0228
p=0.276
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

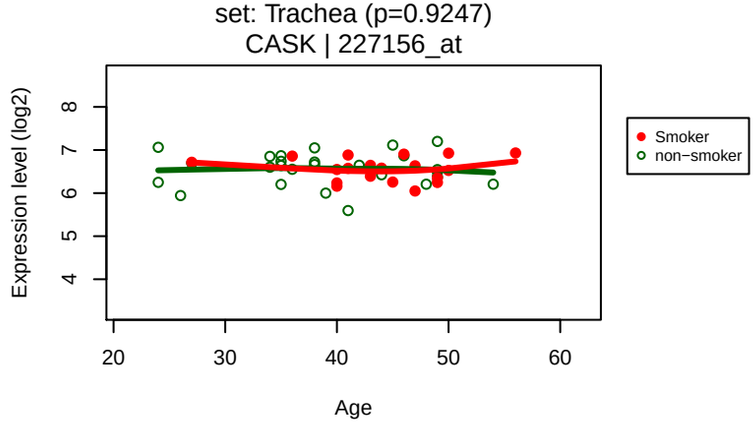

Gene expression ratio in Males
gene: CASK | 227156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1666
p=0.293
p=0.9206
p=0.4379
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASK | 227156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1392
p=0.0241
p=0.7422
p=0.5799
SmallAirway2
LargeAirway2
Trachea
Macrophage2

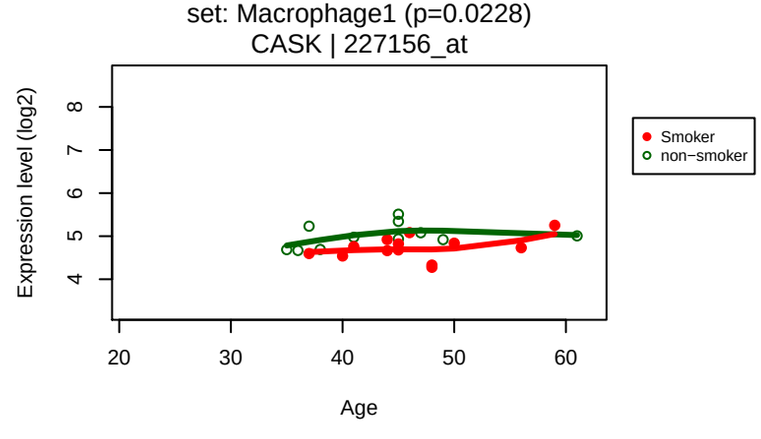

set: SmallAirway1 (p=0.0122)
CASK | 227156_at
Expression level (log2)
Age
Smoker
non−smoker

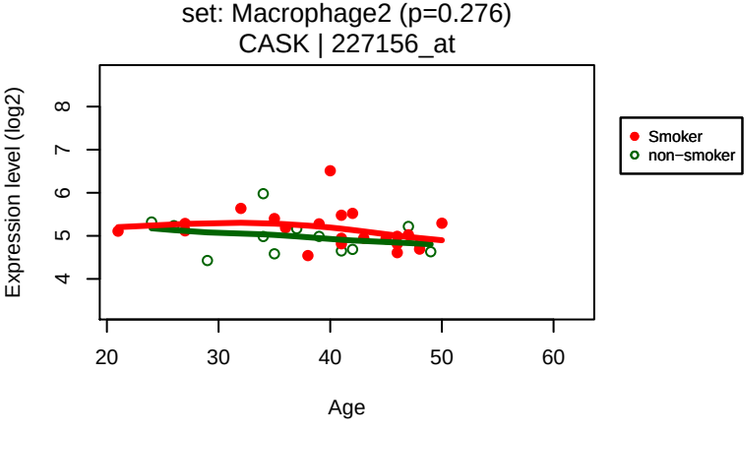

set: SmallAirway2 (p=0.0215)
CASK | 227156_at
Expression level (log2)
Age
Smoker
non−smoker

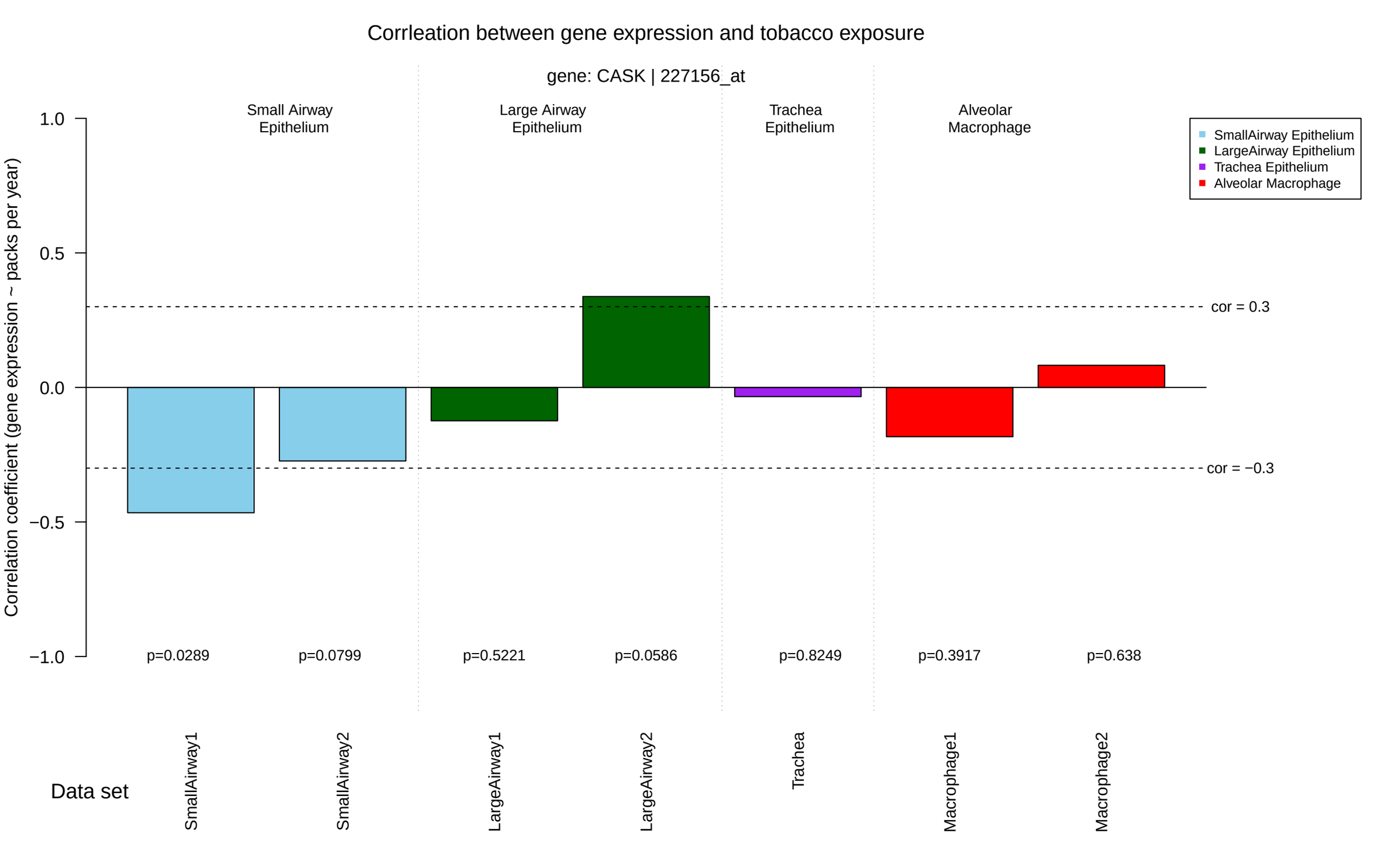

set: LargeAirway1 (p=0.0064)
CASK | 227156_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7783)
CASK | 227156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9247)
CASK | 227156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0228)
CASK | 227156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.276)
CASK | 227156_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASK | 227156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0289
p=0.0799
p=0.5221
p=0.0586
p=0.8249
p=0.3917
p=0.638
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2